

CHE 435/525 *Process Systems Analysis and Control*

Lecture 20: **Stability Margins and Robust Stability**

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Recapture: Bode stability criterion

Two characteristic frequencies for G_{OL}

For the open-loop TF $G_{OL}(s)$, define

- **Gain crossover frequency** ω_g : where $AR = |G_{OL}(i\omega_g)| = 1$;
- **Critical frequency** ω_c (or **phase crossover frequency** ω_p): where $\phi = \angle G_{OL}(i\omega_c) = -\pi$.

Bode stability criterion

Assume that G_{OL} is proper and stable (i.e., the system is **open-loop stable**), with unique ω_g and ω_c :

$$|G_{OL}(i\omega_g)| = 1, \quad \angle G_{OL}(i\omega_c) = -\pi.$$

Then the closed-loop system is stable if either one of the following holds:

$$\phi_g = \angle G_{OL}(i\omega_g) > -\pi, \quad AR_c = |G_{OL}(i\omega_c)| < 1.$$

Learning objectives of this lecture

- ① Calculate **phase margin** and **gain margin** for given feedback loops
- ② Explain the concept of **robust stability**, its difference from **nominal stability**, and its relation to stability margins
- ③ Design controller parameters for **stability margin specifications**

Stability margins

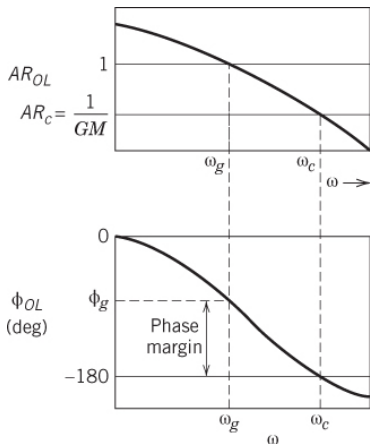
Definitions

Assume that G_{OL} is proper and stable (i.e., the system is **open-loop stable**), with unique ω_g and ω_c :

$$AR_g = |G_{OL}(i\omega_g)| = 1, \quad \phi_c = \angle G_{OL}(i\omega_c) = -\pi.$$

- **Phase margin** $PM = \angle G_{OL}(i\omega_g) - (-\pi) = \phi_g + \pi$
- **Gain margin** $GM = 1/AR_c = 1/|G_{OL}(i\omega_c)|$
- For closed-loop stability, we need $PM > 0$ and $GM > 1 = 0 \text{ dB}$... so that G_{OL} is different from -1 at all frequencies

Stability margins on the Bode diagram of G_{OL}



- **Rule of thumb:** A **well-tuned** controller should make $PM > \pi/6 = 30^\circ$ and $GM > 1.7 = 4.6 \text{ dB}$... so that G_{OL} is **much** different from -1 at all frequencies

Stability margins: Example 1

Calculate GM

Given $G_v = G_m = 1$, $G_p = \frac{K_p}{s(\tau_p s + 1)^2}$, $G_c = K_c$ (P-controller), find GM.

① $G_{OL}(s) = \frac{K_p K_c}{s(\tau_p s + 1)^2},$

$$\text{AR} = \frac{K_p K_c}{\omega(1 + \omega^2 \tau_p^2)}, \quad \phi = -\frac{\pi}{2} - 2 \arctan \omega \tau_p$$

② Find ω_c :

$$\begin{aligned} \phi_c = -\pi &\Rightarrow -\frac{\pi}{2} - 2 \arctan \omega_c \tau_p = -\pi \\ &\Rightarrow \omega_c \tau_p = \tan(\pi/4) \Rightarrow \omega_c = 1/\tau_p \end{aligned}$$

③ Determine GM:

$$\text{AR}_c = \frac{K_p K_c}{\omega_c(1 + 1)} = \frac{1}{2} K_p K_c \tau_p \Rightarrow \text{GM} = 2/K_p K_c \tau_p$$

• Add-on question: Design K_c to achieve $\text{GM} \geq 6 \text{ dB} = 2$.

Stability margins: Example 1

Design for a PM specification

Given $G_v = G_m = 1$, $G_p = \frac{K_p}{s(\tau_p s + 1)^2}$, $G_c = K_c$ (P-controller), design K_c such that $\text{PM} = 30^\circ$.

① $G_{OL}(s) = \frac{K_p K_c}{s(\tau_p s + 1)^2}$,

$$\text{AR} = \frac{K_p K_c}{\omega(1 + \omega^2 \tau_p^2)}, \quad \phi = -\frac{\pi}{2} - 2 \arctan \omega \tau_p$$

② Given $\text{PM} = 30^\circ$, we know that $\phi_g = -150^\circ$:

$$-90^\circ - 2 \arctan \omega_g \tau_p = -150^\circ \Rightarrow \omega_g = \tan 30^\circ / \tau_p$$

③ AR_g must be equal to 1:

$$\frac{K_p K_c}{(\tan 30^\circ / \tau_p)(1 + \tan^2 30^\circ)} = \frac{3\sqrt{3}}{4} K_p K_c \tau_p = 1 \Rightarrow K_c = \frac{4}{3\sqrt{3} K_p \tau_p}$$

Stability margins: Example 2

Calculate PM

Given $G_v = G_m = 1$, $G_p = \frac{K_p}{(\tau_p s + 1)^2} e^{-\theta s}$, $G_c = K_c$ (P-controller), determine PM.

① $G_{OL}(s) = \frac{K_p K_c}{(\tau_p s + 1)^2} e^{-\theta s},$

$$\text{AR} = \frac{K_p K_c}{1 + \omega^2 \tau_p^2}, \quad \phi = -\omega \theta - 2 \arctan \omega \tau_p$$

② Find ω_g :

$$1 = \frac{K_p K_c}{1 + \omega_g^2 \tau_p^2} \Rightarrow \omega_g = \frac{1}{\tau_p} \sqrt{K_p K_c - 1}$$

③ Determine GM:

$$\phi_g = -\frac{\theta}{\tau_p} \sqrt{K_p K_c - 1} - 2 \arctan \sqrt{K_p K_c - 1} \Rightarrow \text{PM} = \phi_g + \pi$$

Stability margins: Example 2

Design for a GM specification

Given $G_v = G_m = 1$, $G_p = \frac{K_p}{(\tau_p s + 1)^2} e^{-\theta s}$, $G_c = K_c$ (P-controller), design K_c such that $\text{GM} = 2$.

① $G_{OL}(s) = \frac{K_p K_c}{(\tau_p s + 1)^2} e^{-\theta s},$

$$\text{AR} = \frac{K_p K_c}{1 + \omega^2 \tau_p^2}, \quad \phi = -\omega \theta - 2 \arctan \omega \tau_p$$

② Given $\text{GM} = 2$, we know that $\text{AR}_c = 1/2$:

$$\frac{K_p K_c}{1 + \omega_c^2 \tau_p^2} = \frac{1}{2} \Rightarrow \omega_c = \frac{1}{\tau_p} \sqrt{2K_p K_c - 1}$$

③ ϕ_c must be equal to -180° :

$$-(\theta/\tau_p) \sqrt{2K_p K_c - 1} - 2 \arctan \sqrt{2K_p K_c - 1} = -\pi$$

④ Numerical solution is needed for $x = \sqrt{2K_p K_c - 1}$ first:

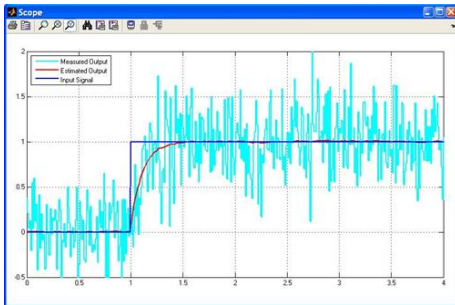
$$(\theta/\tau_p)x + 2 \arctan x = \pi$$

Numerical evaluation of stability margins

- Well... We don't always have closed-form expressions for margins.
- Matlab: `[GM, PM, omega_c, omega_g] = margin(G_OL);`
- Python: `GM, PM, omega_c, omega_g = control.margin(G_OL)`
- Example: $G_{OL} = \frac{2}{(s+1)^2} e^{-s}$
 - ▶ Verify that $\omega_g = 1$, $PM = (\pi/2 - 1) \text{ rad} = 32.72^\circ$
 - ▶ Verify that $\omega_c (= 1.3065)$ satisfies $\omega_c + 2 \arctan \omega_c = \pi$,
 $GM = (1 + \omega_c^2)/2 = 1.3535$

Model uncertainties

- The need for margins arises from **model uncertainties**



- Dude got this data from step response ...
- Maybe the process is $G_p = \frac{K_p}{\tau_p s + 1} e^{-\theta_p s}$ with $K_p = 1$, $\tau_p = 0.5$, and $\theta_p = 1$... **Are we sure about these values?**
- G_c that achieves closed-loop stability with the “model” may not be good with the “actual plant”

Implications of stability margins

- Phase margin

- ▶ Suppose that the actual system has an **extra unknown delay** $\Delta\theta$ than the model

$$G_{OL} = \tilde{G}_{OL}e^{-(\Delta\theta)s}$$

- ▶ G_{OL} & $\tilde{G}_{OL}$ always have equal AR, so ... ω_g doesn't change, but

$$\phi_g = \tilde{\phi}_g - \omega_g \Delta\theta$$

- ▶ In order that $\phi_g > \pi$, $\omega_g \Delta\theta < \text{PM}$... **Maximum allowed unmodeled delay**

$$\Delta\theta_{\max} = \text{PM}/\omega_g$$

- Gain margin

- ▶ Suppose that the actual system has an **extra unknown gain** than the model

$$G_{OL} = K'\tilde{G}_{OL}$$

- ▶ G_{OL} & $\tilde{G}_{OL}$ always have equal ϕ , so ... ω_c doesn't change, but

$$\text{AR}_c = K'\tilde{\text{AR}}_c$$

- ▶ **Maximum allowed unmodeled gain**

$$K'_{\max} = \text{GM}$$

Robust stability

- PM and GM – Capability to tolerate **model uncertainty** or **plant-model mismatch**
- **Nominal stability**: Stable based on a “nominal model” of the plant
 - ▶ $PM > 0$ & $GM > 0$ dB
- **Robust stability**: Stable against possible model uncertainty
 - ▶ PM quite large, GM quite large

** More generic description of robust stability

- **Plant-model mismatch:** $G_{OL}(s) = (1 + \Delta(s))\tilde{G}_{OL}(s)$
 - ▶ $\Delta(s)$: Relative (multiplicative) mismatch
 - ▶ Assume that $|\Delta(i\omega)| \leq \ell(\omega)$: Percentage uncertainty on frequency response at ω
 - ▶ Total upper bound of uncertainty:

$$|G_{OL}(i\omega) - \tilde{G}_{OL}(i\omega)| \leq \ell(\omega)|\tilde{G}_{OL}(i\omega)|$$

- **Robust stability:** At all ω , $\tilde{G}_{OL}(i\omega)$ must keep away from -1 with a distance at least $\ell(\omega)|\tilde{G}_{OL}(i\omega)|$
 - ▶ $|\tilde{G}_{OL}(i\omega) - (-1)| \geq \ell(\omega)|\tilde{G}_{OL}(i\omega)|$

** More generic description of robust stability

Doyle-Stein Theorem (1980)

Suppose the open-loop t.f. is inaccurately modeled, with **relative mismatch bounded in frequency domain**:

$$G_{OL}(s) = (1 + \Delta(s))\tilde{G}_{OL}(s), \quad |\Delta(i\omega)| \leq \ell(\omega).$$

(In other words, the mismatch is uncertain but a bound on its A.R. is known.) Then a sufficient condition for robust stability is

$$\ell(\omega) \left| \frac{G_{OL}}{1 + G_{OL}}(i\omega) \right| < 1 \text{ for all } \omega.$$

- If $G_m = K_m$, $\frac{G_{OL}}{1+G_{OL}} = T = QG_pG_vG_m$ is the t.f. from Y_{sp} to Y !
- Often we can **overestimate** $\ell(\omega)$ by the A.R. of another t.f. $W_T(s)$:

$$\ell(\omega) \leq |W_T(i\omega)|.$$

Then the robust stability can be guaranteed, as long as

$$|(W_T T)(i\omega)| < 1 \text{ for all } \omega.$$